# Data resources for the DDLS PhD project “Uncertainty-aware clinical and MRI models for multiple sclerosis disease-course classification and disability worsening”

November 2026

## 1. Purpose and scope

This document describes the four data sources used in the project, how they are allocated across the four aims, and the main characteristics of each source. All figures are aggregate counts from published papers, manuscripts, approved applications and the data providers’ own documentation. No individual-level data are included. Access to each source is governed by its own ethics approvals and data access agreements.

## 2. Overview of data sources

The project combines two clinical registries, one pooled clinical-trial resource with MRI, and two Swedish clinical trials (Table 1). The Swedish MS Registry (SMSReg) is the main development source for the clinical models, and the MS Clinical and Imaging Data Resource (MS-CIDR) is the only source with MRI.

**Table 1.** Data sources used in the project.

| Source | Type | Size | Modality | Role |
|---|---|---|---|---|
| SMSReg [1, 2] neuroreg.se | National registry, Sweden | 24,441 people, 228,192 visits (2025 extract) [8] | Clinical | Development, Aims 1 and 2 |
| MSBase [3] msbase.org | International registry | 56,578 people, 542,632 visits, 37 countries after curation [9] | Clinical | External evaluation only |
| MS-CIDR [6] progressivemsalliance.org | Pooled clinical trials (14 groups) | 13,074 participants, 199,108 clinical visits, 51,978 MRI visits | Clinical, MRI | External clinical test, Aims 1 and 2; development, Aims 3 and 4 |
| RIFUND-MS and RIDOSE-MS [4, 5] | Phase 3 trials, Sweden | ~325 participants, ~2,200 visits, six-year study period | Clinical | External evaluation, Aims 1 and 2 |

Disease-course labels differ between sources. In SMSReg and MSBase the disease course is recorded by the treating clinician at each visit. In MS-CIDR it is the subtype at trial entry (7,251 RRMS, 3,454 SPMS and 2,369 PPMS participants) and should not be read as an observed RRMS-to-SPMS transition. RIFUND-MS and RIDOSE-MS include RRMS, RIFUND-MS also included clinically isolated syndrome (CIS), and neither trial contains SPMS. The RRMS/SPMS and confirmed disability worsening (CDW) models use only RRMS and SPMS patients. PPMS data from SMSReg and MS-CIDR are used only for self-supervised pretraining.

## 3. Allocation of data across aims

**Table 2.** Use of each source by aim. Aims 1 and 2 develop clinical models; Aims 3 and 4 develop MRI and clinical-MRI fusion models.

| Source | Aim 1 (RRMS/SPMS) | Aim 2 (CDW) | Aim 3 (RRMS/SPMS) | Aim 4 (CDW) |
|---|---|---|---|---|
| SMSReg | Development, internal test, chronology sensitivity | Development, internal test, chronology sensitivity | Not used | Not used |
| MSBase | External evaluation by country | External evaluation, subject to additional permission | Not used | Not used |
| MS-CIDR | External clinical test | External clinical test | MRI and fusion development | MRI and fusion development, using the Aim 3 partitions |
| RIFUND-MS, RIDOSE-MS | External evaluation | External evaluation where event counts permit | Clinical pretraining after Aims 1 and 2 | Clinical pretraining after Aims 1 and 2 |

Training, calibration and test sets are split at patient level in every source, and SMSReg participants linked to RIFUND-MS or RIDOSE-MS are excluded from development in Aims 1 and 2. All models use probability calibration and conformal prediction [7], sequential drift monitoring and integrated gradients. CDW is defined

following Müller et al. [10] in all sources. Models will be deployed on SciLifeLab Serve, where our earlier tool MSP-Tracker is hosted [7].

## 4. Swedish MS Registry (SMSReg)

SMSReg is the nationwide Swedish quality registry for MS and is part of the Swedish Neuro Registries [1, 2]. Two extracts have been used in our previous work (Table 3). Quality control removed duplicates and records missing EDSS, date of birth, disease course at the visit or onset date [8]. The 45,077 visits from 15,719 individuals in SMSReg-2025 that were not part of SMSReg-2022 model development were used for prospective validation [8].

**Table 3.** SMSReg extracts and curated datasets.

| Dataset | Individuals | Visits | Source |
|---|---|---|---|
| SMSReg-2022, extracted 19 December 2022 | 22,748 | 197,227 | [7, 8] |
| SMSReg-2025, extracted 21 February 2025 | 24,441 | 228,192 | [8] |
| SMSReg-2022 after quality control, PPMS excluded | 14,382 | 113,784 | [8] |
| SMSReg-2025 after quality control, PPMS excluded | 19,004 | 158,722 | [8] |
| SMSReg-2025, not used in SMSReg-2022 model development | 15,719 | 45,077 | [8] |
| SMSReg after HARMONISE curation | 15,579 | 127,720 | [9] |

The nine variables harmonised with MSBase in previous work were age at diagnosis, EDSS score, age at visit, sex, year of diagnosis and four treatment categories: no treatment, first-line disease-modifying therapy, second-line disease-modifying therapy and haematopoietic stem cell transplantation [9]. The studies were approved by the Swedish Ethical Review Authority (Dnr 2021-00702, 2024-07442-02 and 2025-00541-02). Data are provided without direct identifiers and the researchers have no access to the re-identification key.

## 5. MSBase

MSBase is an international observational registry for collaborative outcomes research in MS [3]. In previous work, the initial extract contained 109,283 people with MS and 1,308,607 hospital visits from 45 countries. After retaining visits with EDSS, age at visit, sex and a recorded diagnosis of RRMS or SPMS, 56,578 people and 542,632 visits from 37 countries remained [9].

In this project MSBase is used only for external RRMS/SPMS evaluation, reported by country. It is not used for training, fine-tuning, self-supervised pretraining or country-fold training, and earlier cross-country training results are reported as background only. CDW evaluation in MSBase requires additional permission, which is being sought. MSBase is registered with the WHO International Clinical Trials Registry Platform (ACTRN12605000455662) and was approved by the Melbourne Health Human Research Ethics Committee and site institutional review boards, unless exemptions were granted under local regulations. Enrolled patients gave written informed consent. Tomas Kalincik is the MSBase collaborator.

## 6. MS Clinical and Imaging Data Resource (MS-CIDR)

MS-CIDR is a resource of the International Progressive MS Alliance that provides anonymised longitudinal clinical follow-up and conventional MRI from MS clinical trials [6]. Table 4 lists the trials in the resource as reported by the Alliance in its April 2026 data access guidelines. These totals describe the whole resource; the dataset approved for a project may be a subset. MRI visits count imaging time points rather than unique participants or image sequences.

**Table 4.** Clinical trials in MS-CIDR. Extension studies are grouped with their parent study. SPI2 includes 414 SPMS and 228 PPMS participants.

| Trial group | Subtype | Sponsor | Registration | PubMed | Partici-pants | Clinical visits | MRI visits |
|---|---|---|---|---|---|---|---|
| ADVANCE / ATTAIN | RRMS | Biogen | NCT00906399 | 24794721 30181778 | 1,512 | 26,719 | 8,719 |
| CARE-MS I | RRMS | Genzyme, a Sanofi company | NCT00530348 | 23122652 | 581 | 6,538 | 1,685 |
| CARE-MS II | RRMS | Genzyme, a Sanofi company | NCT00548405 | 23122650 | 840 | 9,278 | 2,380 |
| CONFIRM / ENDORSE | RRMS | Biogen | NCT00451451 NCT00835770 | 22992072 27207449 | 1,429 | 25,482 | 4,210 |
| DEFINE / ENDORSE | RRMS | Biogen | NCT00420212 NCT00835770 | 22992073 27207449 | 1,234 | 22,346 | 3,452 |
| OPERA1 | RRMS | Hoffmann-La Roche | NCT01247324 | 28002679 | 820 | 10,925 | 3,344 |
| OPERA2 | RRMS | Hoffmann-La Roche | NCT01412333 | 28002679 | 835 | 10,809 | 3,269 |
| ASCEND-1 / ASCEND-2 | SPMS | Biogen | NCT01416181 | 29545067 | 889 | 28,169 | 5,017 |
| EXPAND | SPMS | Novartis Pharmaceuticals | NCT01665144 | 29576505 | 1,645 | 17,130 | 7,157 |
| MAESTRO-03 | SPMS | BioMS Technology Corp. | NCT00468611 | None | 506 | 3,152 | 1,909 |
| INFORMS | PPMS | Novartis Pharmaceuticals | NCT00731692 | 26827074 | 970 | 13,931 | 3,419 |
| OLYMPUS | PPMS | Genentech, Inc. | NCT00087529 | 19847908 | 439 | 5,267 | 1,970 |
| ORATORIO | PPMS | Hoffmann-La Roche | NCT01194570 | 28002688 | 732 | 13,029 | 3,292 |
| SPI2 | SPMS and PPMS | MedDay Pharmaceuticals SA | NCT02936037 | 33222767 | 642 | 6,333 | 2,155 |
| Total | | | | | 13,074 | 199,108 | 51,978 |

The planned clinical variables are demographics, disease duration, attack history, EDSS and functional systems, Timed 25-Foot Walk, 9-Hole Peg Test, PASAT or SDMT, comorbidities and visit trajectories, where available. The planned MRI sequences are T1 before and after gadolinium, T2, PD, FLAIR, MPRAGE or 3D FLASH, with associated metadata. Trial, site and scanner are used for stratified validation and sensitivity analyses, not as primary predictors.

MS-CIDR is the external clinical test set in Aims 1 and 2 and the development set for the MRI and fusion models in Aims 3 and 4; Aim 4 uses the partitions defined in Aim 3. PPMS participants are used only for self-supervised pretraining. Data are stored on access-restricted university infrastructure, limited to named users under the Data Access Agreement, not redistributed and not linked to external individual-level datasets.

## 7. RIFUND-MS and RIDOSE-MS

RIFUND-MS (NCT02746744) was a multicentre, rater-blinded phase 3 trial comparing rituximab with dimethyl fumarate in RRMS or CIS at 17 Swedish hospitals [4]. RIDOSE-MS (NCT03979456; EudraCT 2018-000721-31) is a randomised phase 3 trial, sponsored by Karolinska Institutet, comparing rituximab 500 mg every 6 months with every 12 months in RRMS [5]. Together the trials provide approximately 325 participants and 2,200 visits over a six-year study period. Ethics approvals are 2016/473-32 (RIFUND-MS) and 2018/1061-31 and 2026-00482-02 (RIDOSE-MS).

The trials serve as external clinical evaluation data in Aims 1 and 2, including CDW where event counts permit. Only after these evaluations are the data reused for clinical pretraining in Aims 3 and 4. Anders Svenningsson is the collaborator for both trials.

## 8. Supervision

The project is supervised by Joachim Burman (main supervisor), with Kim Kultima, Ola Spjuth and Akshai Parakkal Sreenivasan as co-supervisors.

## References


1. Andersen O. From the Gothenburg cohort to the Swedish multiple sclerosis registry. Acta Neurol Scand. 2012;126(Suppl 195):13–19. doi:10.1111/ane.12023
2. Hillert J, Stawiarz L. The Swedish MS registry: clinical support tool and scientific resource. Acta Neurol Scand. 2015;132(Suppl 199):11–19. doi:10.1111/ane.12425
3. Butzkueven H, Chapman J, Cristiano E, et al. MSBase: an international, online registry and platform for collaborative outcomes research in multiple sclerosis. Mult Scler. 2006;12(6):769–774. doi:10.1177/1352458506070775
4. Svenningsson A, Frisell T, Burman J, et al. Safety and efficacy of rituximab versus dimethyl fumarate in patients with relapsing-remitting multiple sclerosis or clinically isolated syndrome in Sweden: a rater-blinded, phase 3, randomised controlled trial. Lancet Neurol. 2022;21(8):693–703. doi:10.1016/S1474-4422(22)00209-5
5. RItuximab Long-Term DOSE Trial in Multiple Sclerosis (RIDOSE-MS). ClinicalTrials.gov identifier NCT03979456.
6. International Progressive MS Alliance. MS Clinical and Imaging Data Resource. Data access guidelines, April 2026. www.progressivemsalliance.org/ms-clinical-and-imaging-data-resource
7. Sreenivasan AP, Vaivade A, Noui Y, Khoonsari PE, Burman J, Spjuth O, Kultima K. Conformal prediction enables disease course prediction and allows individualized diagnostic uncertainty in multiple sclerosis. NPJ Digit Med. 2025;8:224. doi:10.1038/s41746-025-01616-z
8. Sreenivasan AP, Noui Y, Erngren I, Spjuth O, Burman J, Kultima K. Prospective validation of an uncertainty-aware machine learning model for secondary progressive multiple sclerosis classification. Manuscript.
9. Sreenivasan AP, et al. HARMONISE: an uncertainty-aware framework for harmonising disease-course predictions across international clinical registries. Manuscript.
10. Müller J, Sharmin S, Lorscheider J, et al. Standardized definition of progression independent of relapse activity (PIRA) in relapsing-remitting multiple sclerosis. JAMA Neurol. 2025;82(6):614–625. doi:10.1001/jamaneurol.2025.0495